Spectral-Seeded NSGA-II for Multi-Application Mapping in Many-Core Systems

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***Abstract*—Mapping multiple applications onto a Network-on-Chip (NoC) is a complex optimization problem. While evolutionary algorithms like NSGA-II are widely used, their reliance on random initialization causes slow convergence and sub-optimal routing. To address this, we propose a Spectral-Seeded NSGA-II approach. By applying spectral graph partitioning and Fiedler vector analysis, we dynamically cluster highly communicative tasks. Injecting these optimized clusters as elite seeds into the NSGA-II initial population drastically reduces the search space. Preliminary projections indicate that this methodology can accelerate convergence% while improving communication latency, energy efficiency, and thermal balance across the many-core system.**

***Index Terms*—Network-on-Chip (NoC), Application Mapping, NSGA-II, Spectral Clustering, Multiprocessor System-on-Chip (MPSoC).**

I. Introduction

In the era of modern computing, Multiprocessor Systems-on-Chip (MPSoC) and many-core architectures have become the standard for handling intensive, highly parallel workloads. As the number of processing cores integrated into a single chip scales into the hundreds, traditional bus-based communication architectures suffer from severe scalability and performance bottlenecks. Network-on-Chip (NoC) has emerged as the de facto communication paradigm to address these limitations, offering superior scalability, modularity, and high parallel bandwidth. However, the overall efficiency of an NoC-based many-core system heavily relies on how the diverse tasks of concurrent applications are assigned to the available processing cores. This process, known as application mapping, is critical because poor mapping leads to long communication distances, increased latency, thermal hotspots, and excessive energy consumption.

The multi-application mapping problem in NoC is highly complex and has been mathematically proven to be NP-hard. System designers are tasked with simultaneously optimizing multiple conflicting objectives, such as minimizing communication cost, reducing end-to-end latency, and maximizing fault tolerance or energy efficiency. To tackle this, Multi-Objective Evolutionary Algorithms (MOEAs), particularly the Non-dominated Sorting Genetic Algorithm II (NSGA-II), are frequently employed due to their ability to generate a Pareto-optimal front of trade-off solutions. Despite their widespread use, standard MOEAs suffer from a fundamental flaw: they typically initialize their search populations randomly. In the massive, highly constrained search spaces of multi-application NoC mapping, random initialization wastes significant computational resources evaluating poor-quality configurations. Consequently, traditional NSGA-II approaches exhibit slow convergence rates and are highly susceptible to becoming trapped in local optima. While alternative constructive heuristics, such as pure spectral partitioning, can rapidly find topologically sound mappings, they are static and lack the evolutionary mechanisms required to optimize multiple dynamic metrics. Thus, there is a clear gap in the literature for an approach that bridges the mathematical rigor of structural graph partitioning with the multi-objective optimization power of MOEAs.

To address this gap, this paper proposes a novel hybrid optimization framework: *Spectral-Seeded NSGA-II for Multi-Application Mapping in Many-Core Systems*. Instead of relying on a blind, random start, we implement a spectral partitioning and clustering algorithm as a pre-processing stage. Using spectral graph theory, the algorithm analyzes tightly coupled communication patterns within application task graphs and maps them to the NoC topology to minimize the communication cut-size. This topologically aware mapping is then used to intelligently "seed" the initial population of the NSGA-II algorithm. The primary contribution of this work is the integration of deterministic spectral graph partitioning with a stochastic evolutionary search. By initiating the NSGA-II search space from high-quality, communication-optimized individuals, we aim to drastically accelerate the convergence rate. Ultimately, the proposed methodology is expected to yield a broader and more dominant Pareto-optimal front—specifically balancing latency, energy, and fault-tolerance—compared to traditional unseeded metaheuristics in complex multi-application scenarios.

II. Related Work

The task mapping problem in Network-on-Chip (NoC) architectures has been extensively studied, transitioning from exact mathematical models to more scalable heuristic and metaheuristic approaches [1]. Recent literature demonstrates a significant paradigm shift towards integrating machine learning, specifically clustering techniques, into metaheuristic algorithms to establish a "cognitive base" that accelerates convergence and avoids local optima [2].

A. Basic Machine Learning-Assisted Metaheuristics

Several studies have adopted foundational clustering techniques to guide the initial population of evolutionary algorithms. Amin et al. proposed the iHPSA algorithm, which combines Particle Swarm Optimization (PSO) and Simulated Annealing (SA) [2]. By utilizing K-Means clustering equipped with the Elbow method for dynamic cluster count prediction, iHPSA effectively groups high-bandwidth tasks, leading to improved communication costs and energy consumption [2]. Similarly, Sikandar et al. introduced the Sailfish Optimization Algorithm (SFOA), which leverages a Shared K-Nearest Neighbor (KNN) clustering approach to minimize power dissipation and optimize thermal distribution [3]. While these methods achieve rapid initial convergence, they are primarily limited to single-application mapping scenarios.

B. Advanced Predictive and Density-Based Approaches

To further reduce computational overhead, recent works have integrated advanced predictive models. Amin et al. presented GNoC and its extended version GNoC-PR, based on Grey Wolf Optimization [4]. GNoC-PR is notable for incorporating Polynomial Regression to intelligently predict optimal search sub-spaces, reducing computational overhead by up to 40% [4]. On the other hand, Mehmood et al. proposed the Andean Condor Algorithm (ACA), which employs a communication density-based clustering method [5]. ACA dynamically tunes its exploration and exploitation rates based on population fitness, achieving significant savings in computation time while evaluating exact mesh configurations [5].

C. Spectral Clustering-Enhanced Approaches

Addressing the structural complexities of task graphs, Li et al. introduced the ET algorithm (Enhanced Coati Optimization), which utilizes Spectral Clustering validated by the Silhouette Coefficient [1]. This approach captures the structural similarities between tasks more deeply than standard K-Means. Coupled with an incremental hill-climbing strategy, ET achieves highly competitive communication costs and average network latency [1].

Furthermore, purely constructive approaches have leveraged spectral graph theory to achieve rapid and high-quality mappings without the overhead of evolutionary searches. For instance, Alagarsamy et al. proposed SMA, a constructive spectral partitioning method that significantly improves initial mapping speed and communication bandwidth [6]. Similarly, Raj et al. introduced SpecMap, demonstrating the efficacy of spectral partitioning for static application mapping. SpecMap accurately groups highly communicating tasks and eliminates interference, proving highly effective even in multi-task environments [7]. While these algorithms rapidly find topologically sound configurations, they lack the dynamic mechanisms required to simultaneously optimize multiple conflicting objectives like thermal distribution and fault tolerance.

D. Positioning of This Work

Despite the advancements in clustering-based metaheuristics, existing literature predominantly focuses on single-application mapping and relies heavily on single-objective cost functions. In modern Many-Core systems, the simultaneous execution of multiple applications is a realistic constraint [4]. Recent efforts by de Almeida et al. emphasize the absolute necessity of multi-objective algorithms for multi-application mappings, specifically targeting critical metrics such as fault tolerance and thermal distribution [8].

This work distinguishes itself by proposing a Spectral-Seeded NSGA-II model specifically designed for *multi-application* mapping. Unlike prior methods that optimize only communication cost, our multi-objective approach concurrently optimizes latency, communication cost, and thermal distribution, aligning perfectly with the complex requirements of modern MPSoCs. Furthermore, by converting compound task graphs into a unified Affinity Matrix and performing Fiedler vector analysis [9], we inject mathematically profound "elite seeds" into the NSGA-II initialization phase. This novel integration solves the slow convergence issue inherent in massive, multi-application search spaces, directly addressing the limitations of both pure spectral methods and randomly initialized MOEAs [10].

A summary of the key differences between this work and existing methodologies is presented in Table I.

TABLE I
COMPARISON OF CLUSTERING-BASED AND MULTI-OBJECTIVE APPLICATION MAPPING ALGORITHMS

Work	Year	Clustering Strategy	Dynamic K Method	Thermal Aware	Multi-App Support	Algorithm
Sikandar et al. [3]	2021	Shared KNN	Empirical/Static	✓		SFOA
Amin et al. [2]	2022	K-Means	Elbow Method			iHPSA
Alagarsamy et al. [6]	2022	Spectral	-			SMA
Mehmood et al. [5]	2022	Density-Based	Deterministic			ACA
Raj et al. [7]	2023	Spectral	-		✓	SpecMap
Amin et al. [4]	2023	K-Means + PR	Silhouette			GNoC-PR
Li et al. [1]	2025	Spectral	Silhouette			ET
de Almeida et al. [8]	2025	-	-	✓	✓	NSGA-II
This Work	2026	Spectral (Fiedler)	Silhouette	✓	✓	NSGA-II

III. PROBLEM DEFINITION

The multi-application mapping problem in Many-Core Network-on-Chip (NoC) systems involves assigning a set of concurrent application tasks to physical processing cores. Because the number of possible mapping combinations grows factorially with the number of tasks and cores, this is classified as an NP-hard problem. This section formally defines the system architecture, the multi-objective goals, and the constraints governing the mapping process.

A. System / Application Description

The system context involves simultaneously mapping multiple incoming applications onto a Many-Core System or Multiprocessor System-on-Chip (MPSoC) based on a 2D-Mesh Network-on-Chip (NoC) architecture. In this scenario, application behaviors are abstracted as a set of Directed Acyclic Graphs (DAGs), referred to as *Core Graphs*. In these graphs, each node represents a computational task, and each directed edge denotes the communication bandwidth requirement (i.e., data volume) between dependent tasks.

Before the evolutionary mapping process begins, individual task graphs from multiple applications are consolidated into a unified *Affinity Matrix*. The target hardware platform consists of routing nodes, physical communication links, and processing cores operating under a deterministic XY routing algorithm, ensuring a deadlock-free communication environment. The mapping is assumed to be static (design-time), meaning task profiles and communication requirements are known a priori and processed by the Spectral Partitioning module before being passed to the NSGA-II initialization phase.

B. Objectives

The proposed Spectral-Seeded NSGA-II methodology aims to solve a multi-objective optimization problem. The primary objectives are:

- Minimize Communication Cost: Reduce the total physical distance that data packets travel across the NoC, thereby mitigating network congestion and lowering overall energy consumption.
- Minimize Average Network Latency: Decrease the data transfer time between cores by grouping highly communicative tasks within the same topological clusters.
- Optimize Thermal Distribution: Prevent thermal hotspots by avoiding dense placement of high-activity cores in localized regions, thus improving hardware reliability.
- Accelerate Convergence Speed: Reduce the NP-hard search space using spectrally optimized "elite seeds" instead of purely random initialization, thereby avoiding local optima.

C. Constraints

To ensure physical realizability and validity of the generated mappings, the optimization is subject to the following constraints:

- Mapping Exclusivity: Each IP core can host at most one task (1:1 mapping assumption).
- Complete Assignment: Every task must be assigned to exactly one processing core.
- Topological and Architectural Limits: The mapping must fit within the predefined $X \times Y$ dimensions and total core count of the 2D-Mesh NoC.

D. Mathematical Formulation

The multi-objective mapping process requires a formal mathematical representation of both the application set and the target architecture. Let the merged multi-application structure be denoted as a task graph $G(T, E)$ and the target 2D-Mesh architecture as $A(C, L)$.

Decision Variables: The primary decision variable is binary and determines whether a task is assigned to a specific processing core:

$$x_{i,u} = \begin{cases} 1, & \text{if task } t_i \text{ is mapped to core } c_u \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

Objective Functions (Multi-Objective Formulation): The optimization concurrently minimizes two conflicting objectives.

Objective 1: Communication Cost (f_1) Minimizes the total communication distance and bandwidth product:

$$\min f_1 = \sum_{i=1}^{N} \sum_{j=1}^{N} \sum_{u=1}^{M} \sum_{v=1}^{M} x_{i,u} \cdot x_{j,v} \cdot w_{i,j} \cdot H_{u,v} \tag{2}$$

Objective 2: Thermal & Fault-Tolerance Cost (f_2) Promotes uniform thermal distribution by maximizing the distance between active cores, or equivalently, minimizing the distance of active cores to the nearest idle core:

$$\min f_2 = \sum_{u=1}^{M} \left(x_{*,u} \cdot \min_{v \in C_{\text{idle}}} H_{u,v} \right) \tag{3}$$

where $x_{*,u} = 1$ if core u is active, and C_{idle} is the set of unassigned cores.

Constraints:

a) Task Assignment Constraint: Each task must be assigned to exactly one core:

$$\sum_{u=1}^{M} x_{i,u} = 1 \quad \forall i \in \{1, 2, \ldots, N\} \tag{4}$$

b) Core Capacity Constraint: Each core can execute at most one task:

$$\sum_{i=1}^{N} x_{i,u} \leq 1 \quad \forall u \in \{1, 2, \ldots, M\} \tag{5}$$

Symbol	Description
$G(T, E)$	Unified multi-application task graph (DAG)
T	Set of tasks $\{t_1, t_2, \ldots, t_N\}$
E	Set of directed edges representing task dependencies
$w_{i,j}$	Communication volume from task t_i to t_j
$A(C, L)$	NoC architecture graph
C	Set of processing cores $\{c_1, c_2, \ldots, c_M\}$
L	Set of communication links
$H_{u,v}$	Manhattan distance (hop count) between cores c_u and c_v
$x_{i,u}$	Binary mapping decision variable

TABLE II
SUMMARY OF MATHEMATICAL SYMBOLS

IV. System Architecture and Design

This section details the system model and computational architecture of the proposed Spectral-Seeded NSGA-II framework. As an Optimization Track project, the focus lies in mathematically abstracting application characteristics and hardware constraints for efficient mapping optimization rather than physical hardware implementation.

The overall execution flow of the system is illustrated in Figure 1. The architecture operates through two primary stages: First, in the *Pre-processing Stage*, incoming multi-application

task graphs are parsed and consolidated into a unified Affinity Matrix. A Spectral Pre-processor then performs Laplacian and Fiedler vector analysis, utilizing the Silhouette Coefficient for dynamic clustering to identify highly communicative task groups. Second, in the *Evolutionary Stage*, these optimized task clusters are injected as "Elite Seeds" alongside standard random individuals to form a robust initial population. The NSGA-II algorithm then takes over, iteratively applying genetic operators (crossover, mutation) and non-dominated sorting to evolve the solutions. This continuous loop ultimately outputs a Pareto-optimal set of mappings tailored for 2D-Mesh NoC architectures.

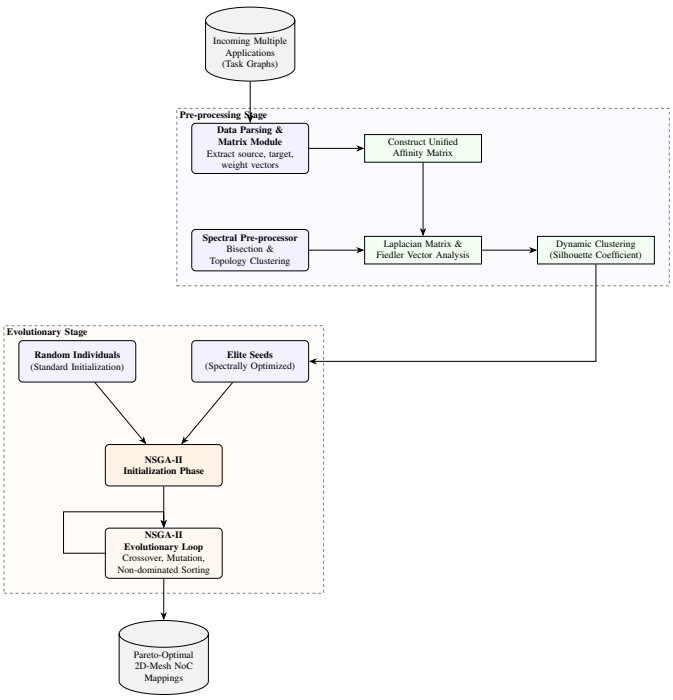

Fig. 1. Detailed System Architecture of the Spectral-Seeded NSGA-II Mapping Model.

A. Hardware/Software Architecture

The target hardware platform is a 9×9 2D-Mesh Network-on-Chip (NoC) architecture, consisting of 81 processing cores arranged in a grid topology. Each core can execute a single task under the 1:1 mapping constraint. Communication between cores follows the XY routing protocol, which ensures deadlock-free packet routing by first routing packets horizontally along the X-axis, then vertically along the Y-axis. The Manhattan distance between any two cores (i, j) and (i', j') is computed as:

$$H_{(i,j),(i',j')} = |i - i'| + |j - j'| \tag{6}$$

This metric is directly related to the physical hop count in the NoC under XY routing.

The optimization pipeline is driven by a specialized software framework designed to abstract the physical NoC environment and execute the metaheuristic search. The core mathematical modeling, including the NoC architectural simulation, matrix operations, and the Spectral Pre-processing phase, is implemented using MATLAB. For the evolutionary optimization stage, the system utilizes PlatEMO—a comprehensive, open-source MATLAB platform dedicated to evolutionary multi-objective optimization. PlatEMO specifically handles the execution of the NSGA-II algorithm.

B. Dataflow and System Model

The dataflow in this project represents how application information traverses the optimization pipeline to become a physical hardware mapping.

The process begins with four independent Directed Acyclic Graphs (DAGs) representing the application workloads. These graphs are parsed and consolidated into a unified block-diagonal *Affinity Matrix*. This matrix is passed to the Spectral Pre-processor, which calculates the Laplacian matrix and performs eigendecomposition. The resulting Fiedler vectors are clustered using the Silhouette Coefficient to group tasks with heavy communication bandwidth requirements.

These topologic clusters (data chunks) are then injected into the NSGA-II algorithm as "Elite Seeds." The evolutionary loop iteratively processes these seeds alongside random individuals, applying crossover and mutation operators, and evaluating them against the hardware NoC constraints until a Pareto-optimal front of mapping solutions is generated.

C. Requirements Analysis

- Data Abstraction: The system must parse multiple application task graphs (DAGs) and merge them into a single Unified Affinity Matrix.
- Autonomous Clustering: The pipeline must compute the Laplacian matrix and dynamically determine the optimal number of topological task clusters using the Silhouette Coefficient.
- Heuristic Seeding & Convergence: The resulting clusters must be injected as "Elite Seeds" into the NSGA-II initial population to significantly accelerate Pareto front convergence compared to pure random initialization.
- Multi-Objective Evaluation: The algorithm must simultaneously minimize communication cost (reducing latency and power) and maximize spatial task distribution (preventing thermal hotspots).
- Hardware Constraints: The genetic operators must strictly maintain a valid 1:1 task-to-core mapping for the target 2D-Mesh NoC architecture.

D. Timing and Execution Model

The system's execution is divided into two distinct phases:

- Offline Phase (Design-Time): Optimization processes (task profiling, clustering, and evolution) run entirely prior to hardware deployment. The main timing constraint is computational efficiency, aiming for algorithm convergence in a small time frame compared to the other models like ILP.

- Runtime Phase (Deployment): Once mapped to the NoC, the system operates on a strictly *data-driven* model. Tasks remain idle until their required input packets arrive, with communication latencies remaining deterministic due to the pre-optimized XY routing.

E. Model of Computation

The system operates under the *Dataflow Model of Computation (MoC)*, where execution is strictly driven by data availability. This aligns perfectly with our application abstraction:

- Task-Level Parallelism: A task executes independently as soon as all its input data dependencies are satisfied.
- Acyclic Structure: The modeled applications (VOPD, MPEG4, VCE, WIFIRX) are Directed Acyclic Graphs (DAGs), ensuring deadlock-free execution and eventual termination.
- Token-Based Interaction: Inter-core communication relies on packetized tokens routed via the deterministic XY network, directly mirroring the weighted edges in our optimization pipeline.

F. Assumptions and Design Limitations

The optimization framework is built upon several critical assumptions:

- Static Mapping: Task-to-core assignments are strictly determined at design-time. Dynamic runtime migration or load balancing is not considered in this model.
- Homogeneous Cores: All 81 processing cores in the NoC are assumed to be identical in computational capacity, operating frequency, and power consumption profiles.
- Known Application Profiles: Task execution times and communication data volumes are constant, known a priori, and do not fluctuate dynamically during runtime.
- No Bandwidth Adjudication: The model minimizes overall communication cost but assumes the NoC physical links possess sufficient bandwidth capacity (i.e., absolute link congestion limits are relaxed).
- 1:1 Mapping Only: Time-multiplexing is excluded. A core executes at most one assigned task.

G. Resource Model

This section outlines the estimated resource footprint of the optimization software for mapping N application tasks onto the 9×9 (81-core) NoC architecture, where $N \leq 81$.

- Memory Usage: The system maintains a highly efficient memory footprint. The sparse Affinity Matrix scales linearly with the number of tasks and communication edges ($O(N + E)$). The precomputed 81×81 Manhattan distance lookup table for the target NoC consumes approximately 52 KB, and maintaining an NSGA-II population of 100 mapping arrays utilizes less than 1 MB of working RAM per generation.
- Computational Overhead: The one-time Laplacian eigendecomposition complexity is $O(N^3)$ at the start of the pipeline, which remains computationally trivial for any task count up to the 81-core limit. Subsequently, the NSGA-II loop executes multi-objective non-dominated sorting with a time complexity of $O(P^2 \cdot M)$ per generation (where P is population size and M is the number of objectives).
- Search Space Reduction: The massive search space of $81!/(81 - N)!$ combinations is drastically reduced by spectrally clustering communicating tasks upfront, saving significant computation time by avoiding millions of poor mapping choices.

V. Methodology

This section details the implementation of the proposed Spectral-Seeded NSGA-II framework. The methodology bridges deterministic graph partitioning algorithms and stochastic multi-objective evolutionary search to address the NP-hard multi-application mapping problem [8]

A. Optimization Pipeline (Optimization Track)

The proposed solution comprises a four-stage optimization pipeline designed to simultaneously process and map a heterogeneous workload of independent applications. The pipeline stages are executed sequentially as follows:

1) Stage 1: Input and Graph Consolidation: To process multiple independent applications through a single spectral clustering pass, their individual Directed Acyclic Graphs (DAGs) are ingested and merged into a unified, block-diagonal *Affinity Matrix* (A) [8]

$$A_{i,j} = \begin{cases} w_{i,j}, & \text{if } t_i, t_j \in T_k \text{ (intra-application edge)} \\ 0, & \text{if } t_i \in T_k, t_j \in T_m, k \neq m \text{ (inter-application)} \end{cases} \tag{7}$$

This block-diagonal structure ensures that subsequent partitioning naturally identifies the boundaries between independent applications while optimizing heavy communication links within them, eliminating inter-application interference during the mapping phase [7]

2) Stage 2: Spectral Graph Partitioning (Processing): Purely random initialization in massive search spaces often leads to slow convergence and sub-optimal routing [2], [4]

The eigenvalue decomposition of L yields eigenvectors where the Fiedler vector (the second smallest eigenvector) captures the fundamental bipartition of the communication graph. By applying K-Means clustering to the eigenspace, tasks are partitioned into k groups that minimize the communication cut. To avoid manual parameter tuning, the optimal number of clusters k is determined dynamically using the Silhouette Coefficient, a technique recently validated in state-of-the-art NoC metaheuristics [1]

3) Stage 3: Elite Seeding and Optimization Method: The optimization is driven by the Non-dominated Sorting Genetic Algorithm II (NSGA-II). Instead of a purely random initialization, the spectral clusters identified in Stage 2 are mapped to physical NoC regions using a dynamic region allocation heuristic. This ensures that high-bandwidth tasks are placed in spatial proximity [3], [5]

These highly optimized mappings are injected into the initial NSGA-II population as "Elite Seeds" (comprising 20% of the population), while the remaining 80% is randomly generated to maintain genetic diversity. The population is then evolved over 100 generations. The chromosome is encoded as a 1D permutation array to enforce a strict 1:1 task-to-core mapping. Partially Mapped Crossover (PMX) and Swap Mutation operators are utilized to traverse the search space without violating hardware constraints.

4) Stage 4: Output: The evolutionary loop continuously evaluates the population against two conflicting objectives: minimizing the total communication cost (f_1) and maximizing the thermal-aware distribution of active cores (f_2). The final output of the pipeline is a Pareto-optimal front of 2D-Mesh NoC mappings, providing system designers with diverse trade-off configurations.

B. Algorithm Description

The Spectral-Seeded NSGA-II framework consists of four main stages designed to solve the multi-application mapping problem. The following sections provide the step-by-step pseudocode for each stage of the process.

Algorithm 1: Graph Consolidation and Block-Diagonal Affinity Matrix

Require: m independent application DAGs $\{G_1, \ldots, G_m\}$; Communication edge weights
Ensure: Block-diagonal Affinity Matrix $A \in \mathbb{R}^{N \times N}$ with $N = \sum |T_i|$
Initialize $A \leftarrow \mathbf{0}_{N \times N}$
offset $\leftarrow 0$
for each application G_k **do**
  for each edge $(t_i, t_j) \in G_k$ **do**
    $A[\text{offset} + i, \text{offset} + j] \leftarrow w_{i,j}$
  end for
  offset $\leftarrow$ offset $+|T_k|$
end for
return A

Algorithm 2: Spectral Clustering with Silhouette Optimization

Require: Affinity Matrix A
Ensure: Optimal clusters $\{C_1^*, \ldots, C_{k^*}^*\}$; Optimal count k^*
Compute Laplacian $L \leftarrow D - A$ where $D = \text{diag}(A\mathbf{1})$
$(V, \Lambda) \leftarrow \text{Eigendecomp}(L)$ ▷ Extract Fiedler vectors
Normalize rows of V to unit norm
for $k = 2$ to $K_{\max}$ **do**
  $S_k \leftarrow \text{Silhouette}(\text{KMeans}(V[:, 1:k], k))$
end for
$k^* \leftarrow \arg\max_k S_k$
$\{C_1^*, \ldots, C_{k^*}^*\} \leftarrow \text{KMeans}(V[:, 1:k^*], k^*)$
return Clusters, k^*

Algorithm 3: Dynamic Region Allocation and Elite Seed Generation

Require: Spectral clusters $\{C_1, \ldots, C_{k^*}\}$; NoC mesh $X \times Y$
Ensure: Initial population P_0 with 20% elite seeds, 80% random
Elite seeds:
for $i = 1$ to $0.2 \times N_{\text{pop}}$ **do**
  Allocate each cluster to spiral-fill mesh region
  Greedily place tasks to minimize intra-cluster communication
  Add elite chromosome π to P_0
end for
Random chromosomes:
for $i = 1$ to $0.8 \times N_{\text{pop}}$ **do**
  Generate random valid permutation π_{rand}
  Add to P_0
end for
return P_0

Algorithm 4: NSGA-II with Early Stopping Criteria

Require: Population P_0; Max generations $G_{\max} = 100$; Stagnation window $w = 10$
Ensure: Pareto-optimal front F^* of task-to-core mappings
$P \leftarrow P_0$, stag_count $\leftarrow 0$
while gen $< G_{\max}$ AND stag_count $< w$ **do**
  Evaluate: Compute f_1 (communication cost) and f_2 (thermal balance) for all $\pi \in P$
  Sort: Non-dominated sorting and crowding distance
  Select: Tournament selection to create parent pool Q
  Reproduce: PMX crossover ($p_c = 0.9$) + Swap mutation ($p_m = 1/N$) to create offspring Q'
  Merge: Environmental selection: $P \leftarrow \text{EliteSelect}(P \cup Q', N_{\text{pop}})$
  Check: If $\text{HyperVolume}(F^{\text{curr}}) \approx \text{HyperVolume}(F^{\text{prev}})$, then stag_count++
end while
return Final Pareto front from P

VI. Experimental Setup

A. Platform and Tools

All simulations, including graph processing and evolutionary optimization, are executed in MATLAB R2023b using the **PlatEMO v2.8** framework [10] on a standard desktop workstation.

B. Datasets and Workloads

Based on the multi-application evaluation methodology proposed by de Almeida et al. [8], four standard NoC benchmarks (*VOPD, MPEG4, VCE, and WIFIRX*) are merged to construct a unified heavy workload. This approach results in a single comprehensive dataset consisting of 73 computational tasks and 111 communication edges, which are mapped concurrently onto the target 9×9 (81-core) NoC architecture.

C. Evaluation Metrics

The proposed Spectral-Seeded NSGA-II is benchmarked against a standard random-initialized NSGA-II and a pure static spectral mapping. The evaluation relies on the following metrics:

- **Convergence Speed:** The number of generations required for the Pareto front to stabilize.
- **Communication Cost (f_1):** The bandwidth-weighted Manhattan distance, representing network latency and dynamic power consumption.
- **Thermal Balance (f_2):** The spatial distribution of active cores, ensuring heat dissipation and hardware reliability.
- **Hypervolume (HV):** A standard multi-objective metric evaluating both the accuracy (convergence) and diversity (spread) of the final Pareto front.

VII. System-Level Considerations

This section discusses critical system-level design aspects across the optimization lifecycle. While some features are partially implemented, others are identified as future enhancements.

A. Latency and Real-Time Constraints

- Communication Latency Minimization: The optimization directly minimizes weighted Manhattan distances, reducing task-to-task communication time. Spectral clustering ensures nearby tasks are co-located, naturally satisfying real-time latency constraints.
- Deterministic Routing: The XY routing protocol guarantees deadlock-free, predictable path lengths between any two cores. This ensures bounded, analyzable latency.

B. Reliability and Fault Tolerance

- Thermal Distribution Objective: The f_2 objective minimizes the distance to spare cores, distributing heat and enabling fast failure recovery.
- Spare Core Locality: By maximizing proximity of active tasks to idle cores, the mapping supports rapid task relocation in case of core failure.

C. Energy Efficiency

- Communication Energy Reduction: By minimizing hop counts (via f_1), the framework reduces energy dissipated in network links, which often dominates total chip power.
- Thermal-Aware Placement: Objective f_2 promotes thermal balance, reducing cooling requirements and overall energy footprint in data centers.

D. Scalability

The proposed algorithm scales to larger systems as follows:

- Spectral Clustering: Eigendecomposition cost is $O(n^3)$ where n is the task count. For $n \leq 1000$ tasks, this remains tractable on modern CPUs.
- NSGA-II: Non-dominated sorting scales as $O(N^2M)$ where N is population size and M is objective count. Population size can grow linearly with task count.

E. Algorithm Dynamics: Seeding Impact on Solution Diversity

A critical consideration in initializing metaheuristics with elite seeds is the risk of premature convergence. If the initial population is overly dominated by spectral clusters, NSGA-II may lose genetic diversity and become trapped in a local optimum dictated by the Fiedler vector.

To mitigate this, the proposed system employs a *Partial Seeding Strategy*. As defined in Section V, only 20% (N_{elite}) of the initial population is seeded using the spectral assignments. The remaining 80% is generated completely randomly. Furthermore, NSGA-II's built-in *Crowding Distance* metric actively penalizes overly dense solution clusters in the objective space, ensuring that the evolutionary search explores the trade-off boundaries rather than purely exploiting the communication-optimized spectral seeds.

Acknowledgments

The proposed framework combines deterministic spectral partitioning with stochastic evolutionary optimization. The formulation of the Affinity Matrix and the use of the Fiedler vector for spectral clustering are adapted from Todorov et al. [9], the SpecMap approach by Raj et al., and SMA by Alagarsamy et al. [6]. The optimization phase is based on NSGA-II, following its successful application in multi-objective NoC mapping by de Almeida et al. [8].

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